Amol Prakash

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Austin, Texas | Open to Relocate | Open to Remote

Summary

Healthcare AI & Research Data Analyst with an **M.S. in Health Informatics**, an **M.B.B.S.** (foreign MD), **PMI CAPM–certified project professional**, and **3+ years** of experience developing machine learning–enabled healthcare analytics, integrating large-scale **EHR**, **REDCap**, registry, and administrative datasets, and building reproducible data pipelines for clinical research. Skilled in **SQL**, **Python**, **R**, and **SAS**, with applied experience in cloud-based healthcare analytics, predictive modeling, and healthcare AI supporting cancer prevention, population health, and federally funded research.

Education

Master of Science, Health Informatics Aug 2023 – May 2025
Indiana University, Indianapolis — GPA: 3.8/4.0

- **Relevant Coursework:** Healthcare Data Science, Biostatistics, Health Informatics & Analytics, Machine Learning & AI in Healthcare, Data Visualization, Project Management

M.B.B.S, Doctor of Medicine Sep 2011 – Jun 2017
Uzhhorod National University, Ukraine — GPA: 3.5/4.0

- **Relevant Coursework:** Internal Medicine, Clinical Medicine, Public Health, Epidemiology, Healthcare Administration

Experience

Research Data Analyst Jun 2025 – Present
School of Health & Human Sciences, Indiana University School of Medicine *Remote – Indianapolis, IN*

- Collected, integrated, and managed large-scale clinical and research data from **Cerner EHR (PowerChart), REDCap, NIH All of Us, NHANES, operational systems, and survey datasets** to develop analysis-ready cohorts supporting observational studies, predictive analytics, and healthcare AI research.
- Performed comprehensive data cleaning, preprocessing, feature engineering, validation, and quality assurance across **500,000+ participant records** using **SQL, Python, R and Google BigQuery**, preparing structured datasets for statistical modeling and machine learning analyses.
- Developed and evaluated supervised **machine learning models** using **Python (Scikit-learn, XGBoost)** to investigate clinical outcome prediction and patient risk stratification, comparing model performance through cross-validation and standard evaluation metrics.
- Developed NLP preprocessing workflows using **Python, spaCy, and transformer-based libraries** to extract structured information from clinical narratives and survey responses, supporting cohort identification and downstream machine learning analyses.
- Designed interactive **Power BI and Tableau dashboards**, generated publication-ready statistical reports and visualizations, and collaborated with clinicians, researchers, and data scientists to translate analytical findings into manuscripts, presentations, and healthcare AI research initiatives.

Clinical Research Data Analyst Sep 2024 – May 2025
Department of Pediatrics, Indiana University School of Medicine *Indianapolis, IN*

- Managed longitudinal pediatric HIV clinical datasets across **130+ immune biomarkers**, performing data integration, preprocessing, validation, and quality assurance to support statistical analyses and translational research.
- Developed analysis-ready datasets using **Python, SQL, R, and SPSS**, performing missing data assessment, feature engineering, variable standardization, and cohort construction for longitudinal analyses and predictive modeling.
- Applied supervised **machine learning** methods to investigate predictors of treatment response and adverse clinical outcomes, supporting exploratory analyses and biomarker discovery.
- Designed and optimized **15+ REDCap databases** with branching logic, validation rules, calculated variables, and automated quality checks, reducing data entry errors by **40%** while improving research workflow efficiency.
- Developed publication-quality scientific visualizations using **R** (**ComplexHeatmap**, **ggplot2**, and **pheatmap**), producing clustered heatmaps, longitudinal biomarker plots, correlation matrices, and manuscript-ready figures for NIH-funded pediatric HIV research.

Research Data Analyst Aug 2023 – Dec 2024
Purkayastha Lab for Health Innovation (PLHI), Indiana University *Indianapolis, IN*

- Managed ETL pipelines and integrated healthcare datasets across **200,000+ patient records** using **SQL, MySQL, Python, R, and Google BigQuery**, supporting epidemiologic research and healthcare analytics initiatives.

- Performed advanced statistical analyses and predictive modeling using **Python and R**, including regression, classification, hypothesis testing, and disease surveillance analyses to identify clinical and public health trends.
- Developed and validated **time-series machine learning models** using wearable Fitbit sensor data to classify circadian rhythm disruptions, achieving **87% accuracy** through feature engineering and model optimization.
- Implemented reproducible machine learning workflows using **Python, Scikit-learn, and Git**, incorporating feature engineering, model validation, hyperparameter optimization, and experiment documentation to support healthcare analytics research.

Physician – Investigator Jan 2021 – Jun 2023

NCARF *Kochi, India*

- Directed data coordination for cross-functional research teams of **8+ members** across community-based epidemiologic studies covering **1,200+ households**, improving study efficiency by **25%** from project initiation through completion.
- Conducted field data collection, household surveys, and community outreach in low-resource settings, collaborating with local health officials to translate research objectives into actionable public health interventions.
- Oversaw clinical research data management in **REDCap** and **SAS** for **500+ participants**, ensuring **ICH-GCP** compliance, accurate documentation, data integrity, and audit readiness.
- Developed study timelines, enrollment projections, and risk mitigation strategies, maintaining **98% on-time completion** of research milestones across multiple community-based clinical studies.

Projects

Immune Dysregulation Patterns in Perinatal HIV Exposure | Poster | GitHub Repository Mar 2025

- Served as principal analyst for a pediatric HIV immunology study analyzing **130+ immune markers** across **200+ patients**, applying non-parametric statistical methods and multiple-testing corrections to identify immune dysregulation patterns.
- Designed interpretable visualizations using **z-score standardization**, hierarchical clustering, and heatmaps to evaluate age-dependent immune trajectories across HIV exposure groups.

Hospital Readmission Risk Predictor – Medicare Claims Data | GitHub Repository | Live Demo Feb 2026

- Built an end-to-end machine learning pipeline using CMS DE-SynPUF claims data from **116K+ patients** and **66K+ admissions** to predict 30-day hospital readmission risk.
- Performed feature engineering, class imbalance handling, and model evaluation; deployed models via a **Streamlit application**, achieving **19.4% precision** and **46.4% recall** for high-risk patient identification.

PillPal – Medication Adherence Platform | GitHub Repository | Live Demo Jan 2025

- Developed a **Flutter-based IoT platform** integrating Bluetooth-enabled medication dispensers with real-time adherence tracking and SQLite-based longitudinal data storage.
- Designed predictive dashboards and alert systems to identify at-risk patients up to **72 hours before missed doses**, enabling proactive clinical interventions.

NeoRoo Digital Health Platform | Open Source Contributor Oct 2024

- Contributed to an open-source neonatal monitoring platform using **Flutter** and **Dart**, supporting real-time vital sign tracking and mobile clinical data capture.
- Implemented features for temperature monitoring, visualization dashboards, and notification workflows to improve usability and support neonatal care delivery.

Cardiovascular Disease Risk Stratification | GitHub Repository Aug 2024

- Developed supervised machine learning models to predict cardiovascular disease risk using clinical and demographic data, benchmarking performance across multiple algorithms.
- Applied model validation techniques including cross-validation, ROC-AUC analysis, and feature importance evaluation to ensure model robustness and interpretability.

Certifications

PMI Certified Associate in Project Management (CAPM) — Project Management Institute (PMI)
Data Literacy Certificate — DataCamp
AI Fundamentals certificate — DataCamp
Intermediate SQL — DataCamp
Prompt Engineering and Programming with OpenAI — Columbia+

Publications

Consumer Health Informatics Funding in 2025
Published on ResearchGate, January 2025
ResearchGate Publication Link

Awards & Recognition

BioHealth Informatics Research Grant Nov. 2024
Indiana University School of Medicine *Awarded for AMIA 2024 presentation (San Francisco, CA)*

Luddy Distinguished MS Student Scholarship Aug. 2023 – May 2025
Indiana University *Merit-based full tuition scholarship recognizing academic excellence*

Skills

Programming:	Python, R (tidyverse, ggplot2, Shiny), SQL, SAS, Kotlin, Dart
Libraries:	Pandas, NumPy, SciPy, scikit-learn, Statsmodels, XGBoost, LightGBM, TensorFlow, PyTorch, Keras, Matplotlib, Seaborn, ComplexHeatmap
AI & NLP Tools:	Hugging Face Transformers, LangChain, OpenAI API, spaCy, NLTK, Sentence-BERT, LlamaIndex, Generative AI, Prompt Engineering
Analytics & Visualization:	Tableau (Advanced), Power BI, RStudio, Jupyter Notebook, Google Colab, Excel (Pivot Tables, Power Query), R Shiny, Plotly, D3.js
Statistical Methods:	Regression Modeling, ANOVA, Multivariate Analysis, Survival Analysis, Longitudinal Modeling, Non-parametric Tests, Bayesian Inference, Time-Series Analysis, Meta-analysis
Healthcare Standards:	HL7 FHIR, OMOP CDM, LOINC, SNOMED CT, ICD-10-CM/PCS, CPT, REDCap, HIPAA/HITECH, CDISC (SDTM/ADaM)
ML Techniques:	Supervised & Unsupervised Learning, Deep Learning, Ensemble Methods, Feature Engineering, Cross-Validation, Model Optimization, Explainable AI (SHAP, LIME)
Data Platforms & Tools:	Google BigQuery, MySQL, PostgreSQL, SQL Server, Docker, Apache Spark, Git/GitHub, Android Studio, Flutter, Cloud Integration (Google Cloud, AWS EC2)
Research:	Clinical Trial Management, IRB Protocols, Grant Writing, Manuscript Preparation, Scientific Communication, Collaboration (Slack, Notion)